

ShaileeCart – Full Stack E-Commerce Website

Project Report

Project Overview

ShaileeCart is a full-stack E-Commerce web application developed using React.js, Django REST Framework, PostgreSQL, JWT Authentication, and Tailwind CSS. The platform allows users to browse products, view product details, register/login securely, manage shopping carts, and place orders.

Objectives

- Build a modern full-stack e-commerce platform.
- Implement secure user authentication using JWT.
- Provide product browsing and shopping cart functionality.
- Integrate React frontend with Django REST APIs.
- Store data using PostgreSQL.

Technology Stack

Frontend: React.js, Vite, Tailwind CSS, React Router

Backend: Django, Django REST Framework

Database: PostgreSQL

Authentication: JWT (Simple JWT)

Media Handling: Django Media Files

Version Control: Git & GitHub

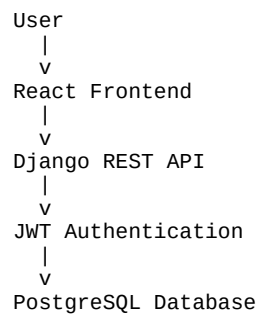
Key Features

- User Registration & Login
- JWT Authentication & Authorization
- Product Listing & Product Details
- Category Management
- Shopping Cart Management
- Quantity Update & Item Removal
- Order Creation & Checkout
- Responsive User Interface

Database Entities

Category, Product, UserProfile, Cart, CartItem, Order, OrderItem, User

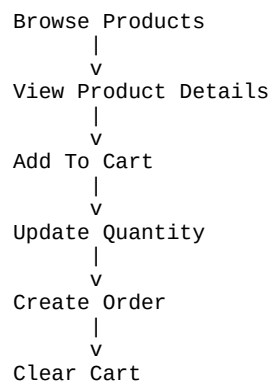
System Workflow



User Authentication Flow



Shopping Cart Flow



API Endpoints

GET /api/products/
GET /api/products/{id}/
GET /api/categories/
GET /api/cart/
POST /api/cart/add/
POST /api/cart/remove/
POST /api/cart/update/
POST /api/orders/create/
POST /api/register/
POST /api/token/
POST /api/token/refresh/

Project Architecture

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graph TD
    Frontend["Frontend (React)"] --> API["REST API Calls"]
    API --> Backend["Backend (Django REST Framework)"]
    Backend --> Logic["Business Logic & Authentication"]
    Logic --> Database["PostgreSQL Database"]
```

Challenges Solved During Development

- Media/image rendering between React and Django.
- JWT authentication integration.
- Cart management and quantity updates.
- API communication and CORS configuration.
- Database relationships and order processing.

Future Enhancements

- Online Payment Gateway Integration
- Product Search & Filtering
- Wishlist Feature
- User Reviews & Ratings
- Admin Analytics Dashboard
- Order Tracking System

Conclusion

ShaileeCart successfully demonstrates the development of a modern full-stack E-Commerce platform using React and Django REST Framework. The project showcases frontend-backend integration, secure authentication, database management, REST APIs, and responsive user interface design.